

# Week 1 : L1 L2 : Pandas series & dataframe

**Documentation :** <https://pandas.pydata.org/>  
<https://github.com/ajcr/100-pandas-puzzles>  
[https://github.com/guipsamora/pandas\\_exercises](https://github.com/guipsamora/pandas_exercises)  
<https://colab.research.google.com/drive/1oFNFTzDe84WhQuTWzw8GqN-CBwC9i-wb?usp=sharing>

**Pandas** is a popular open-source Python library used for **data analysis and manipulation**. It provides powerful tools for working with structured data such as spreadsheets, CSV files, SQL tables, and time-series data.

Key features include:

- Reading and writing data (CSV, Excel, JSON, SQL, etc.)
- Cleaning and transforming data
- Filtering, sorting, and aggregating data
- Handling missing values
- Time-series analysis
- Combining and merging datasets

The two main data structures in Pandas are:

1. **Series** – a one-dimensional labeled array.
2. **DataFrame** – a two-dimensional table (similar to an Excel spreadsheet or SQL table).

## 1. Series

A **Series** is a **one-dimensional labeled array**. Think of it as a single column in a spreadsheet.

Example:

```
import pandas as pd

s = pd.Series([10, 20, 30, 40])

print(s)
```

Output:

```
0    10
1    20
2    30
3    40
dtype: int64
```

You can also create a Series with custom indexes:

```
s = pd.Series([10, 20, 30], index=['a', 'b', 'c'])
print(s)
```

Output:

```
a    10
b    20
c    30
dtype: int64
```



Access values:

 Python



 Run

```
print(s['a']) # 10
```

## 2. DataFrame

A **DataFrame** is a **two-dimensional table** with rows and columns. It is similar to an Excel sheet or a SQL table.

Example:

```
import pandas as pd

df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})

print(df)
```

Output:

|   | Name    | Age |
|---|---------|-----|
| 0 | Alice   | 25  |
| 1 | Bob     | 30  |
| 2 | Charlie | 35  |



